

plete the task (e.g., it’s impossible to definitively say whether this text is offensive without more context), (2) the inability to adopt the persona (e.g., “I cannot answer that question. I am an AI and don’t have personal opinions or beliefs”), (3) a general refusal (e.g., “I’m sorry, but I can’t assist with this request”), (4) refusal of the request due to the inappropriateness of the content (“I cannot provide a label of "offensive" or "not offensive" as it is not appropriate to make subjective judgments about content.”), or (5) a lack of context (e.g., “I cannot classify the text, it is not appropriate for me to make such a determination without more context or information”).

We evaluate the frequency of refusals across models by using a simple string-match method similar to the approach of Röttger et al. (2024b) and de Araujo and Roth (2024). This method classifies as *refusal* any response that starts with “I’m sorry”, “as an AI”, “I cannot”, or similar variations as refusals. See Appendix C for details.

4 Results

4.1 Overall Refusal

Table 1 presents an overview of the variation on false refusals across the different model families and the three tasks we test. In general, there is large variation in the refusal rates across different tasks and models when using persona-based prompting.

In the following sections, we discuss in depth the results for each source of false refusals: task (§4.2), model (§4.3), sociodemographic personas (§4.4), and prompt paraphrases (§4.5).

4.2 Refusal by Task

Here, we ask: **How do false refusals vary across tasks when prompting with personas?** Among the three tasks we evaluate, the offensiveness task has the highest rate of false refusals, with an average of 14.68% across models, followed by politeness (5.64%) and NLI (1.37%) (see Table 1). Politeness shows moderate refusals, and NLI has the lowest refusal rates.

Beyond overall refusal rates, we find that the variability in refusals also depends on the task. The offensiveness task shows the widest range, with refusal rates varying between 0% and 87.36% across different models. Politeness also has a notable range, ranging from 0% to 35.69%, while NLI exhibits the most consistent behavior, with refusal rates varying from 0% to 12.56%. This pattern shows a big difference: tasks that involve sensitive

Model	NLI	Politeness	Offensiv.
Llama2-7B	8.87	30.08	76.54
Llama2-13B	12.56	35.69	87.36
Llama3-8B	0.06	1.59	23.45
Llama3.1-8B	0.04	0.16	6.12
Llama3.2-1B	0.03	0.09	1.90
Llama3.2-3B	0	0	0.10
Qwen1.5-7B	0	0.02	0.39
Qwen1.5-32B	0.15	11.86	17.27
Qwen2-7B	0	0.16	2.07
Qwen2.5-7B	0	0	0.19
Qwen2.5-32B	0	0	0
Gemma-2B	0	0.04	0.18
Gemma-7B	0.08	0.03	0.19
Gemma2-2B	0.07	0.72	2.20
Gemma2-9B	0.05	7.71	13.18
Gemma2-27B	0	2.02	3.80
Mean	1.37	5.64	14.68

Table 1: % of false refusals for each task (NLI, politeness, offensiveness) across models averaged across personas. Horizontal dashed lines separate model families. Offensiv.: Offensiveness.

content (offensiveness and politeness) probably get more refusals, while objective tasks (NLI) probably get fewer refusals because their criteria are clear and logical. Our results suggest that **the task influences model false refusals, with tasks involving sensitive content eliciting an increased number of false refusals compared to objective tasks.**

4.3 Refusal by Model

How do false refusals vary across models when prompting with personas? We test 16 models across 9 different model families, including Llama-2, Llama-3 (and its variants 3.1 and 3.2), Qwen1.5, Qwen2, Qwen2.5, Gemma, and Gemma2 — including a range of small to medium-sized models (1B, 2B, 3B, 7B, 8B, 9B, and 32B). We want to observe how false refusal patterns evolve across and within model families, i.e., whether newer versions improve by reducing false refusal rates.

As shown in Table 1, refusals are restricted to specific models. False refusals in **Llama models** drop substantially from the earlier to the later series. The oldest model in its medium size (Llama2-13B) shows the highest rates (87.36% for offensiveness, 35.69% for politeness and 12.56% for NLI), whereas Llama3-8B shows a substantial decrease (23.45% for offensiveness, 1.59% for politeness

and 0.06% for NLI) yet maintains a high refusal rate. With the Llama3 series, this trend continues since refusal rates for all tasks reduce to almost 0. Most notably, Llama3.2-3B registers no refusals at all. This suggests that later Llama models strategically reduce false refusals to sociodemographic persona prompts.

The **Qwen models** show low false refusals, except for the largest version of the early iteration (Qwen1.5-32B), which has a higher rate in politeness (11.86%) and offensiveness (17.27%), but a low rate in NLI (0.15%). Qwen2 models lowered refusals but still indicated a small amount of false refusal (2.07%) in the offensiveness task. The Qwen2.5 series improves this behavior by reaching near-zero refusals across all tasks, including in its largest model (32B). Similar to the Llama models, the newer Qwen iterations show significant improvements in reducing false refusals.

Unlike Llama and Qwen, the earliest versions of **Gemma models** show low false refusals, but surprisingly, the latest Gemma2 series models have a lot more false refusals. This increase is particularly true for the medium size 9B model, which has a false refusal rate of 7.71% for politeness and 13.18% for offensiveness. Unlike Llama and Qwen, whose newer iterations reduce false refusals, the latest Gemma models show a significant increase.

Thus, **false refusal behavior is more closely tied to model choice, with model scale having a smaller impact**. While newer versions of Llama and Qwen show improvements, false refusals persist with the new generations of Gemma models.

4.4 Refusal by Sociodemographic Personas

We have seen that task choice (§4.2) and model choice (§4.3) strongly impact false refusals. Here, we compare persona-based and persona-free prompting strategies to see if certain personas increase false refusals.

Persona vs. persona-free prompting We analyze how sociodemographic personas influence false refusals by measuring the difference in refusal rates between persona-based and persona-free prompts (§2.3). Given that the offensiveness task gets the highest number of false refusals, we select this task for our analysis.

On average across models, false refusal rates are much higher in the *persona* setup (14.68%). This difference is clearly reflected in Figure 1, which shows greater variation in refusal rates within the

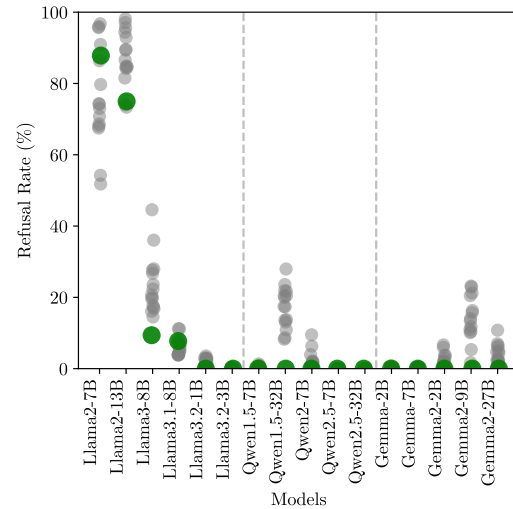


Figure 1: Comparison of refusal rates (%) by model in the offensiveness task across two setups: *persona* (gray) and *persona-free* (green). Vertical dashed lines separate Llama, Qwen and Gemma models.

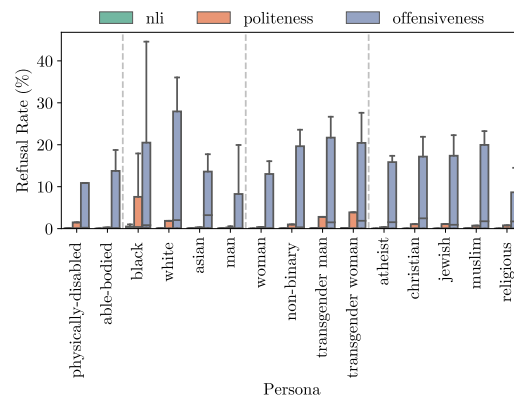


Figure 2: Variation of refusal rates (%) per **persona** across tasks (nli, politeness, offensiveness) aggregated across models. Vertical dashed lines separate sociodemographic groups (disability, race, gender, religion).

persona setup across models. We observe substantial increases in Llama2-13B ($\Delta 12.38$), Llama3-8B ($\Delta 14.08$), Qwen1.5-32B ($\Delta 17.27$), Gemma2-9B ($\Delta 13.15$) and Gemma2-27B ($\Delta 3.78$). Out of 16 models, only six (Llama3.2-3B, Qwen1.5-7B, Qwen2.5-(7B, 32B), and Gemma-(2B, 7B) show no false refusals in both setups. These results clearly indicate that, in most cases, **prompting with sociodemographic personas amplifies false refusals across models**. This effect is especially pronounced in the latest iterations of Gemma2.

False refusal disparities across personas Seeing that persona prompting elicits more false refusals on average, we now investigate whether spe-

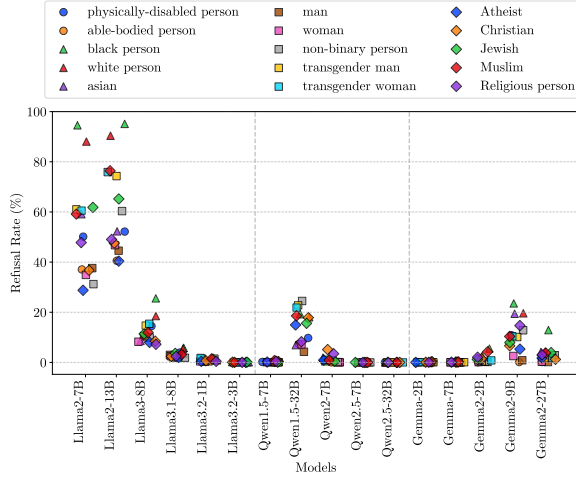


Figure 3: Refusal rates (%) of models across the 15 sociodemographics, averaged over the politeness and offensiveness tasks. Markers indicate sociodemographic categories. Vertical dashed lines separate models.

cific personas elicit this behavior more. Figure 2 shows the variation of false refusals by sociodemographic persona, aggregated across models. We observe 1) that false refusal rates are uneven across sociodemographic personas, and 2) there is significant variability in refusals among models for each persona (e.g., for *black* some models never refuse while some refuse 40% of the time). This is particularly true for the offensiveness task.

Since we see variation across sociodemographic personas, we investigate whether it is systematic at the model level. We compute the refusal rates for the 15 sociodemographic groups, averaging the results over two tasks per model (Figure 3). We find that there is some consistency in which personas explain refusal. Across most models, the top 5 sociodemographics that elicit more refusals are *black*, *white*, *transgender woman*, *transgender man*, and *muslim* personas with an average of 14.67%, 12.34%, 8.43%, 8.28% and 8.33% respectively, across tasks. In the following, we identify some trends: Llama2, Llama3, Llama3.1, and Gemma2 models have high refusal rates for *black* and *white* personas. For *black* person, these Llama series have an average of 47.85% false refusals across tasks, compared to 9.37% for the Gemma2 series. For *white* person, the rates are 41.49% for the Llama models and 5.92% for Gemma2. Offensiveness is the task that triggers more refusals in these sociodemographics across models, as shown in Figure 9 in Appendix D. The largest version (32B) of Qwen1.5 refuses the most for *transgen-*

der man (15.29%), *transgender woman* (14.67%) and *non-binary* (16.33%) personas averaged across tasks, with politeness being the task that triggers more refusals for these sociodemographics (see Figure 8 in Appendix D). Conversely, the top five sociodemographics eliciting the least false refusals are *Christian*, *woman*, *Atheist*, *man*, and *able-bodied person* with an average of 5.62%, 4.53%, 4.49%, 4.32% and 4.24% respectively across models and task. In sum, **we find consistency in the sociodemographics that lead to more false refusals across several models; some groups are more likely to experience false refusals, particularly vulnerable groups based on race, gender, and religion.** This inconsistency reveals underlying biases across sociodemographics in these models and highlights failures in the balance between the safety mechanisms and fairness of these models.

4.5 Refusal by Prompt

Next, we examine the role of prompt paraphrases in shaping false refusals, considering personas. Figure 4 shows variation in false refusals across models and prompt strictness response levels (*unforced-response*, *semi-forced response* and *forced-response*) for the offensiveness task. **A striking finding is that models tend to refuse more when not forced to answer (*unforced-response*), i.e., when prompts are less restrictive and allow broader interpretation.** This trend is particularly evident for several models on the offensiveness task, with refusal rates of 60.92% for Llama2-7b, 74.29% for Llama2-13B, 54.64% for Llama3-8B, 51.98% for Qwen1.5-32B, and 39.65% for Gemma2-9B. The politeness task shows similar trends, though to a lesser degree (see Figure 11 in Appendix D). The NLI task is less affected by false refusals: the prompts exhibit little to no variation (Figure 10 in Appendix D).

4.6 Quantifying Sources of False Refusals

After identifying sources of false refusal, we use statistical methods (a global sensitivity measure and a logistic regression analysis) to *quantify* the impact their impact on refusal behavior.

4.6.1 Wasserstein Distance

We use a global sensitivity measure based on optimal transport (OT), a method from statistics, machine learning, and image processing (Chen et al., 2021). OT quantifies distance between probability measures by finding the minimal-cost